

Market basket Analysis with Python- Amazon

Dataset link:

<https://docs.google.com/spreadsheets/d/1S1ONVMKXjvxEtXzCkl sdXGoeA5BpFTg7/edit?usp=sharing&ouid=117961085555071943983&rtpof=true&sd=true>

Video Link:

<https://drive.google.com/file/d/1gj96kDF-G04wQpqolPReNV80xNDlka7X/view?usp=sharing>

1. What products or categories are frequently purchased together, indicating cross-selling opportunities?

Based on a co-occurrence (market-basket-style) analysis of the **amazon_data_refresh** dataset—where each order can include multiple purchase categories—the following **product categories are most frequently purchased together**, indicating strong **cross-selling opportunities**.

Key Category Pairs Purchased Together

1. **Clothing and Fashion + Others**
 - *Most frequent combination*
 - Indicates apparel buyers often add miscellaneous or accessory-type products to the same order.
2. **Clothing and Fashion + Groceries and Gourmet Food**
 - Suggests lifestyle or convenience-driven purchasing behavior.
 - Opportunity to promote daily-use items alongside fashion deals.
3. **Clothing and Fashion + Home and Kitchen**
 - Shows overlap between apparel shoppers and household product buyers.
 - Effective for seasonal campaigns (e.g., festive shopping, home refresh).
4. **Beauty and Personal Care + Clothing and Fashion**
 - Strong lifestyle synergy.
 - Ideal for bundling fashion items with cosmetics, grooming, or personal care products.
5. **Home and Kitchen + Others**
 - Customers buying home essentials often explore additional categories in the same session.
6. **Groceries and Gourmet Food + Home and Kitchen**
 - A natural complement (e.g., cookware, storage, kitchen tools).

- High potential for recommendation-based upselling.
- 7. **Beauty and Personal Care + Home and Kitchen**
 - Indicates household-centric shoppers with broader consumption needs.
- 8. **Beauty and Personal Care + Groceries and Gourmet Food**
 - Suggests routine, replenishment-based shopping behavior.

Cross-Selling Implications

- **Bundle Recommendations:**
 - Apparel + Beauty
 - Groceries + Kitchen Essentials
- **Personalized Recommendations:**
 - If a customer adds *Clothing and Fashion*, recommend *Beauty*, *Home*, or *Groceries* items.
- **Homepage / Cart Suggestions:**
 - Use “Frequently Bought Together” widgets for the above category pairs.
- **Promotional Strategy:**
 - Cross-category discounts (e.g., “Buy Fashion, Get 10% off Beauty”).

Conclusion

The analysis clearly shows that **Clothing and Fashion**, **Beauty and Personal Care**, **Groceries and Gourmet Food**, and **Home and Kitchen** form a tightly connected purchasing ecosystem. Leveraging these associations through **bundling, recommendations, and targeted promotions** can significantly improve average order value and customer satisfaction.

2. How do customer purchasing patterns vary across demographics or regions?

Based on analysis of the **amazon_data_refresh** dataset, customer purchasing patterns show **clear variation across demographic segments**, particularly by **age group**, **gender**, and **purchase frequency behaviour**. (Regional analysis is not possible because no geographic/region field is present in the dataset.)

1. Purchasing Patterns by Age Group

Younger Customers (18–30 years)

- Purchase more frequently (**weekly or more**).
- Strong preference for:
 - **Clothing and Fashion**
 - **Beauty and Personal Care**
- More exploratory behavior:

- Higher browsing frequency
 - Greater responsiveness to personalized recommendations
- Tend to combine lifestyle categories in a single transaction.

Insight:

This segment is ideal for **trend-driven promotions**, flash sales, and influencer-based recommendations.

Middle-Aged Customers (31–50 years)

- Balanced purchasing behavior.
- Frequently buy:
 - **Groceries and Gourmet Food**
 - **Home and Kitchen**
 - **Beauty and Personal Care**
- Higher cart completion rates.
- More deliberate decision-making, relying on:
 - Reviews
 - Ratings accuracy

Insight:

Best suited for **bundle offers**, subscription-based products, and value-driven cross-selling.

Older Customers (51+ years)

- Lower purchase frequency (**monthly or less**).
- Prefer:
 - **Groceries and Gourmet Food**
 - **Home and Kitchen**
- Less influenced by recommendations or browsing features.
- Higher emphasis on reliability and necessity.

Insight:

Focus on **essential items**, trust signals, and simplified purchasing experiences.

2. Purchasing Patterns by Gender

Male Customers

- Higher likelihood of purchasing:
 - **Electronics-related “Others”**
 - **Home and Kitchen**
 - **Groceries**
- Fewer categories per transaction.
- Lower browsing time; more goal-oriented shopping.

Insight:

Use **direct recommendations** and functionality-focused product suggestions.

Female Customers

- More diverse baskets with multiple categories.
- Strong engagement with:
 - **Clothing and Fashion**
 - **Beauty and Personal Care**
- Higher responsiveness to:
 - Reviews
 - Personalized recommendations
- More frequent cross-category purchases.

Insight:

Excellent target group for **cross-selling**, curated collections, and loyalty programs.

“Prefer Not to Say / Others”

- Mixed behavior, closer to the overall average.
- No strong category dominance, but still shows multi-category purchases.

3. Purchase Frequency and Behavior

- **Frequent buyers (weekly):**
 - More likely to purchase multiple categories together.
 - Less cart abandonment.
- **Infrequent buyers (monthly or less):**
 - Single-category, necessity-based purchases.
 - Higher sensitivity to price and need.

4. Regional Insights (Data Limitation)

- The dataset **does not contain region, city, or country attributes**, so regional variation cannot be quantitatively analyzed.
- If regional data were added, expected differences would likely appear in:
 - Category preference
 - Purchase frequency
 - Delivery satisfaction

Overall Conclusion

Customer purchasing patterns vary significantly across **age and gender demographics**:

- **Younger customers** → lifestyle and trend-driven, high engagement
- **Middle-aged customers** → value-focused, multi-category buyers
- **Older customers** → necessity-driven, low-frequency buyers
- **Female customers** → higher cross-selling potential
- **Male customers** → focused, utility-driven purchases

These insights can directly support **demographic-specific marketing**,

recommendation systems, and cross-selling strategies.

3. Can we segment customers based on buying behaviour to target personalized promotions?

Yes, **customers can be effectively segmented based on buying behavior**, and this segmentation can be directly used to design **highly targeted and personalized promotions**. Using the available dataset, meaningful behavioral segments can be identified without relying on demographic or regional data alone.

1. Behavioral Segmentation Approach

Customers can be segmented using the following behavioral dimensions:

- **Purchase frequency** (weekly, monthly, occasional)
- **Category diversity** (single-category vs multi-category buyers)
- **Browsing behavior** (high vs low browsing frequency)
- **Cart behavior** (completion vs abandonment)
- **Responsiveness to recommendations**
- **Review sensitivity and engagement**

This approach aligns with industry-standard **RFM and behavioral clustering models**.

2. Key Customer Segments Identified

1. Frequent Multi-Category Buyers (High-Value Customers)

Characteristics:

- Purchase weekly or more
- Buy across multiple categories (fashion, beauty, home, groceries)
- High cart completion
- Actively engage with recommendations

Personalized Promotions:

- Exclusive bundle offers
- Loyalty rewards and early-access sales
- Personalized “frequently bought together” suggestions

2. Lifestyle Shoppers

Characteristics:

- Strong preference for **Clothing and Fashion and Beauty**
- High browsing activity
- Influenced by reviews and personalization
- Medium-to-high purchase frequency

Personalized Promotions:

- Trend-based collections
- Influencer-style recommendations
- Cross-category discounts (fashion + beauty)

3. Essential & Utility Buyers

Characteristics:

- Focus on **Groceries** and **Home and Kitchen**
- Lower browsing time
- Monthly or necessity-driven purchases
- Less influenced by recommendations

Personalized Promotions:

- Subscription discounts
- Replenishment reminders
- Bulk-buy and value packs

4. Price-Sensitive Occasional Buyers

Characteristics:

- Low purchase frequency
- Higher cart abandonment
- Limited category exploration
- Sensitive to price and delivery cost

Personalized Promotions:

- Limited-time discounts
- Free shipping offers
- Price-drop alerts

5. Exploratory / Window Shoppers

Characteristics:

- High browsing and search activity
- Low conversion rate
- Often save items for later

Personalized Promotions:

- Personalized retargeting
- Reminder emails and notifications
- Social proof (ratings, reviews, bestsellers)

3. How to Implement This Segmentation

1. Feature Engineering

- Number of categories per transaction
- Purchase frequency score
- Cart completion rate
- Recommendation response score

2. Clustering Techniques

- K-Means or Hierarchical Clustering
- RFM-style scoring

3. Activation

- Map each segment to a promotion strategy

- Deploy via email, app notifications, and homepage personalization

4. Business Impact

- Higher **conversion rates**
- Increased **average order value**
- Improved **customer retention**
- Reduced **promotion cost wastage**

Conclusion

Behavior-based customer segmentation is not only feasible but **highly effective** for personalized promotions. By targeting customers based on **how they shop rather than who they are**, businesses can deliver more relevant offers, improve customer experience, and maximize revenue.

4. Which products drive repeat purchases or high customer loyalty?

Based on behavioral signals available in the **amazon_data_refresh** dataset (purchase frequency, category repetition, cart completion, and satisfaction-related fields), the following **product categories are most strongly associated with repeat purchases and high customer loyalty**. Note that loyalty here is inferred from **behavioral consistency**, not an explicit loyalty flag.

1. Groceries and Gourmet Food – Strongest Driver of Repeat Purchases

Evidence from data:

- Frequently purchased on a **weekly or monthly** basis
- High repetition across transactions
- Low cart abandonment
- Often purchased alongside Home and Kitchen items

Reason:

- Consumable and replenishment-based products naturally drive repeat buying.
- Customers tend to return to the same platform for convenience and reliability.

Loyalty Indicators:

- High purchase frequency
- Routine buying behavior
- Stable satisfaction ratings

Conclusion:

This category is the **primary driver of habitual usage and platform stickiness**.

2. Home and Kitchen – Medium-to-High Loyalty Category

Evidence from data:

- Strong co-occurrence with groceries
- Repeat purchases over time (utilities, storage, kitchen tools)
- Higher cart completion rates

Reason:

- Products are need-based and often upgraded or replaced.
- Trust in product quality and reviews drives repeat decisions.

Loyalty Indicators:

- Consistent category revisits
- Review- and rating-driven purchases

Conclusion:

Home and Kitchen supports **long-term retention**, especially when bundled with essentials.

3. Beauty and Personal Care – Brand and Routine Loyalty

Evidence from data:

- Regular repeat purchases
- High sensitivity to reviews and recommendations
- Often purchased by frequent shoppers

Reason:

- Personal-use products create **brand attachment**
- Customers prefer reordering familiar items once trust is established

Loyalty Indicators:

- Repeat category selection
- Higher engagement with personalized recommendations

Conclusion:

This category drives **brand-level loyalty and repeat engagement**.

4. Clothing and Fashion – Engagement-Driven but Lower Repeat Predictability

Evidence from data:

- High frequency among younger and lifestyle shoppers
- Strong cross-category purchases
- Less predictable repeat timing

Reason:

- Fashion purchases are discretionary and trend-based.
- Repeat purchases depend more on promotions and new arrivals.

Loyalty Indicators:

- High browsing activity
- Engagement-driven rather than necessity-driven

Conclusion:

Clothing drives **engagement loyalty**, not consistent repeat purchasing.

5. “Others” Category – Low Loyalty, Transactional

Evidence from data:

- Purchased frequently with other categories
- Low standalone repeat behavior

Reason:

- One-off or need-specific items
- Low emotional or habitual attachment

Conclusion:

Minimal contribution to long-term loyalty.

Overall Loyalty Ranking (Highest to Lowest)

1. **Groceries and Gourmet Food** – strongest repeat driver
2. **Beauty and Personal Care** – routine and brand loyalty
3. **Home and Kitchen** – functional, trust-based loyalty
4. **Clothing and Fashion** – engagement-focused loyalty
5. **Others** – transactional, low retention impact

Business Implications

- Use **groceries and beauty products** for:
 - Subscriptions
 - Reorder reminders
 - Loyalty programs
- Bundle **home essentials with groceries** to reinforce retention.
- Use **fashion** to attract and engage, then convert loyalty via essentials.

Final Conclusion

Repeat purchases and customer loyalty are primarily driven by consumable and routine-based categories, especially Groceries and Gourmet Food, followed by Beauty and Personal Care and Home and Kitchen. These categories should be the foundation of any loyalty, subscription, or retention strategy.